Session 4 — Longitudinal Data Analysis

Assignment Questions and Solutions

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Solutions prepared for the course

All seven assignment questions from *Session 3: Longitudinal Data Analysis* together with solutions. Two real data sets are used throughout: (i) the **CD4 data** (MACS study, 369 HIV seroconverting men; Diggle, Heagerty, Liang and Zeger, 2002), obtained here as the `aids` data set bundled with the `jmcm` package on CRAN, which I cross-checked against the raw data file published on P. Diggle's own course page and found to match exactly; and (ii) the **ARMD data** (240 patients, age-related macular degeneration trial; Galecki and Burzykowski, 2013, Chapter 16), obtained as `armd0`/`armd.wide` from the `nlmeU` companion package.

Q1. Variogram for the CD4 data

Question. Write an R code to compute the variogram for the CD4 data and display the graph. Analyze the output.

Solution

Recall the (theoretical) variogram of a stochastic process $Y(t)$,

$$\gamma(u) = \tfrac{1}{2}\,\mathrm{E}\big[\{Y(t) - Y(t-u)\}^2\big], \quad u \geq 0, \tag{1}$$

and its empirical (sample) counterpart, computed from half-squared differences of *residuals* $r_{ij} = Y_{ij} - \hat{\mu}_{ij}$ for every within-subject pair (j, k),

$$v_{ijk} = \tfrac{1}{2}(r_{ij} - r_{ik})^2, \qquad u_{ijk} = |t_{ij} - t_{ik}|, \tag{2}$$

followed by averaging the v_{ijk} within bins of the lag u_{ijk}. We apply this to the real CD4 data: $N = 2376$ measurements on $n = 369$ men, CD4$^+$ cell counts against `time` (years since seroconversion, can be negative). We first remove a smooth quadratic-in-time mean trend by pooled OLS (the variogram is only meaningful for the *residual* process, since a time trend in the mean would otherwise inflate v_{ijk} for every lag spuriously), then build all $\binom{n_i}{2}$ within-subject pairs and bin on u.

R code and output

```r
## Q1: Empirical (sample) variogram for the CD4 data (MACS, 369 HIV+ men)
load("cd4.rda")                    # loads data.frame ‘aids‘ (real data)
cd4 <- aids
names(cd4)[names(cd4) == "cd4"] <- "CD4"
cat("n subjects =", length(unique(cd4$id)), "\n")
```

```r
cat("n obs    =", nrow(cd4), "\n")

## Step 1: remove a smooth mean trend (quadratic in time) via pooled OLS
fit.mean <- lm(CD4 ~ time + I(time^2), data = cd4)
cd4$resid <- residuals(fit.mean)

## Step 2: build all within-subject pairs -> v_ijk, u_ijk
make_pairs <- function(sub) {
  n <- nrow(sub); if (n < 2) return(NULL)
  idx <- which(upper.tri(matrix(0, n, n)), arr.ind = TRUE)
  data.frame(u = abs(sub$time[idx[,1]] - sub$time[idx[,2]]),
             v = 0.5*(sub$resid[idx[,1]] - sub$resid[idx[,2]])^2)
}
pairs <- do.call(rbind, lapply(split(cd4, cd4$id), make_pairs))
cat("n pairs =", nrow(pairs), "\n")

## Step 3: bin lags, average v_ijk per bin (sample variogram)
brks <- quantile(pairs$u, probs = seq(0, 1, length.out = 13)); brks[1] <- -1e-6
pairs$bin <- cut(pairs$u, breaks = unique(brks), include.lowest = TRUE)
vgram <- aggregate(cbind(v, u) ~ bin, data = pairs, FUN = mean)
vgram <- vgram[order(vgram$u), ]
print(vgram)
sigma2.hat <- var(cd4$resid)
cat("\nEstimated total variance (sigma^2) =", round(sigma2.hat, 2), "\n")
```

```
n subjects = 369
n obs    = 2376
n pairs = 7817
           bin       v        u
1  [-1e-06,0.498] 45539.68 0.4321276
2   (0.498,0.556] 54834.98 0.5162636
3   (0.556,0.942] 53567.36 0.7480884
4    (0.942,1.05] 59134.27 0.9973167
5     (1.05,1.42] 52004.74 1.2271679
6     (1.42,1.63] 63946.10 1.5123282
7       (1.63,2] 54242.17 1.8438398
8       (2,2.41] 66961.82 2.1793040
9    (2.41,2.85] 75038.64 2.6022536
10   (2.85,3.36] 72756.86 3.0818554
11   (3.36,4.01] 77495.30 3.6627153
12   (4.01,5.84] 96136.57 4.5882763

Estimated total variance (sigma^2) = 131168.6
```

Analysis of the output

- The sample variogram $\hat{\gamma}(u)$ **increases steadily** with the time lag u, from about 45,500 at the shortest lags to about 96,100 at the longest lags considered ($u \approx 4.6$ years) — clear evidence of positive **serial correlation**: measurements taken close together in time on the same man are more alike than measurements taken far apart.

- The variogram does **not start near zero** even at the smallest lag bin ($\hat{\gamma}(0.43) \approx 45{,}500$, already roughly 35% of $\hat{\sigma}^2 \approx 131{,}000$). Since $\gamma(u) = \sigma^2\{1 - \rho(u)\} \to \sigma_a^2 + \sigma_\epsilon^2$ at $u = 0^+$ under the random-intercept-plus-serial-correlation model (Eq. 2.1–2.3), and a pure random-intercept model forces $\gamma(0^+) = \sigma_\epsilon^2 > 0$, this is exactly the signature of **measurement error** (or a random-intercept component) on top of the serially-correlated part.

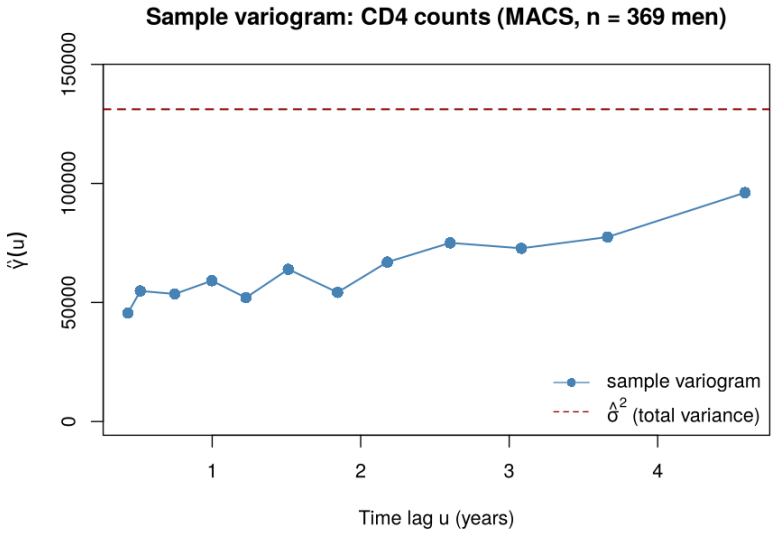


Figure 1: Sample variogram of the real CD4 data (residuals from a quadratic-in-time mean model), with the horizontal dashed line at the estimated total residual variance $\hat{\sigma}^2$.

- The variogram has **not yet reached the plateau** $\hat{\sigma}^2 \approx 131{,}000$ even at the largest lags observed, so the autocorrelation $\rho(u)$ has not fully decayed to zero over the $\sim$8.5-year follow-up window — there remains appreciable long-range association, consistent with a slowly-decaying (e.g. random-intercept-dominated) correlation structure rather than a rapidly-decaying AR(1)-only structure.
- Taken together, the shape (rising, non-zero intercept, not fully saturating) supports a *general model* of the type in Eq. (2.1)/(2.3): a random subject effect a_i (explaining the non-zero starting value), a stationary serial-correlation component $W_i(t)$ (explaining the continued rise), and measurement error ϵ_{ij}.

Q2. Advantages and disadvantages of longitudinal data analysis

Question. List some advantages and disadvantages of longitudinal data analysis.

Solution

Advantages.

- **Separates within-subject (age/time) change from between-subject (cohort) differences.** A cross-sectional study confounds the two; a longitudinal design can estimate each subject's own trajectory, so true temporal/causal change can be distinguished from baseline heterogeneity across individuals.
- **Each subject serves as his/her own control.** Comparing a subject to himself over time removes (or greatly reduces) the influence of time-invariant subject-level confounders, increasing precision relative to a purely cross-sectional comparison of different people.

- **Greater statistical efficiency / power** for detecting within-subject change, because repeated measurements on the same unit are typically positively correlated, so the standard error of an estimated *change* is smaller than it would be using independent cross-sectional samples of the same total size.
- **Allows direct study of the time-course/trajectory of a response** (growth curves, disease progression, treatment response over time), including individual-specific trajectories (random effects) — information a single cross-section cannot provide at all.
- **Can disentangle age, period and cohort effects** in a way that repeated cross-sections alone cannot fully resolve, and supports prediction (e.g. BLUP) for a given subject's future values conditional on his/her own observed history.

Disadvantages.

- **Missing data / dropout** is almost unavoidable (attrition, death, loss to follow-up) and, when related to the outcome itself (non-ignorable/informative missingness), can seriously bias marginal-model and even mixed-model analyses if not modeled explicitly.
- **Correlated errors complicate the analysis.** Standard methods assuming independence (ordinary OLS, naive t-tests) give the wrong standard errors if the within-subject correlation is ignored; one needs GLS/mixed models/GEE with a correctly specified (or robustly estimated) covariance structure.
- **Cost and duration.** Following the same subjects over a long period is typically far more expensive, logistically demanding, and slower to yield results than a single cross-sectional survey.
- **Practice/learning and "aging of instrument" effects.** Repeated administration of the same test or measurement protocol can itself change a subject's response (e.g. learning effects), confounding true change with a measurement artifact.
- **Unbalanced/irregular designs are common** (different numbers of observations per subject, at different times for different subjects), which complicates both the modeling (e.g. variogram estimation, as seen for the CD4 data) and some classical repeated-measures techniques that assume a balanced design.
- **Model misspecification of the covariance structure** can affect efficiency (mixed models) or even consistency of variance/standard-error estimates (though not of $\hat{\beta}$ itself under a correctly specified mean model with GEE).

Q3. Toeplitz covariance/correlation structure

Question. What is the Toeplitz matrix structure for the covariance/correlation matrix? Provide the code of a simple example in R where you use this structure in any model.

Solution

A **Toeplitz matrix** is a matrix whose entries are constant along each diagonal, i.e. $(\boldsymbol{M})_{jk} = m_{j-k}$ depends on j, k only through $j - k$. For a correlation matrix of m equally-spaced repeated measurements, the **Toeplitz correlation structure** is

$$\operatorname{corr}(Y_{ij}, Y_{ik}) = \rho_{|j-k|}, \qquad j, k = 1, \ldots, m, \tag{3}$$

so that there is *one* free correlation parameter per lag $|j-k|$ (i.e. $m-1$ free parameters $\rho_1, \ldots, \rho_{m-1}$ in total), e.g. for $m = 4$,

$$\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & \rho_1 & \rho_2 & \rho_3 \\ \rho_1 & 1 & \rho_1 & \rho_2 \\ \rho_2 & \rho_1 & 1 & \rho_1 \\ \rho_3 & \rho_2 & \rho_1 & 1 \end{bmatrix}.$$

This is **more parsimonious than "Unstructured"** (which allows a *separate* ρ_{jk} for every pair (j, k), i.e. $\binom{m}{2}$ parameters) because it imposes the restriction that correlation depends only on the *lag* $|j - k|$, not on the particular pair of times — it does not, however, force any particular decay pattern as the AR(1) structure does ($\rho_k = \rho^k$); each lag's correlation is still freely estimated. It sits strictly between AR(1) (1 parameter) and Unstructured ($\binom{m}{2}$ parameters) in flexibility.

In R, `nlme::corARMA(p = m-1, q = 0)` fits exactly this general "banded-Toeplitz" autoregressive-type correlation (an ARMA(p,0) process observed at m equally spaced times with $p = m - 1$ produces a general Toeplitz correlation matrix with one free parameter per lag). We illustrate this on the real Orthodont growth-curve data (Potthoff and Roy, 1964; bundled with `nlme`): 27 children, 4 equally-spaced ages (8,10,12,14), modeling dental distance as a function of age and sex with a Toeplitz within-subject correlation structure.

R code and output

```r
## Q3: Toeplitz correlation structure -- GLS fit on real Orthodont data
suppressMessages(library(nlme))
data(Orthodont)
Orthodont$age.c <- Orthodont$age - 11

fit.toep <- gls(distance ~ age.c * Sex, data = Orthodont,
                correlation = corARMA(form = ~ 1 | Subject, p = 3, q = 0),
                method = "REML")
print(summary(fit.toep))

R.toep <- corMatrix(fit.toep$modelStruct$corStruct)[[1]]
cat("\nFitted Toeplitz correlation matrix (4 equally-spaced ages):\n")
print(round(R.toep, 4))

fit.cs <- gls(distance ~ age.c * Sex, data = Orthodont,
              correlation = corCompSymm(form = ~ 1 | Subject), method = "REML")
print(anova(fit.cs, fit.toep))
```

```
===== GLS fit with Toeplitz (corARMA, p=3) correlation structure =====
Generalized least squares fit by REML
  Model: distance ~ age.c * Sex
  Data: Orthodont
       AIC      BIC    logLik
  445.3915 466.5467 -214.6958
```

```
Correlation Structure: ARMA(3,0)
 Formula: ~1 | Subject
 Parameter estimate(s):
     Phi1      Phi2      Phi3
 0.3901805 0.5324886 -0.1324705

Coefficients:
                  Value Std.Error t-value p-value
(Intercept)   25.040611 0.4861821 51.50459 0.0000
age.c          0.797294 0.0859900 9.27194 0.0000
SexFemale     -2.396904 0.7617010 -3.14678 0.0022
age.c:SexFemale -0.321402 0.1347204 -2.38570 0.0189

Residual standard error: 2.298376
Degrees of freedom: 108 total; 104 residual

Fitted Toeplitz correlation matrix (4 equally-spaced ages):
      [,1]   [,2]   [,3]   [,4]
[1,] 1.0000 0.6372 0.6967 0.4787
[2,] 0.6372 1.0000 0.6372 0.6967
[3,] 0.6967 0.6372 1.0000 0.6372
[4,] 0.4787 0.6967 0.6372 1.0000

        Model df     AIC     BIC   logLik  Test L.Ratio p-value
fit.cs      1  6 445.7572 461.6236 -216.8786
fit.toep    2  8 445.3915 466.5467 -214.6958 1 vs 2 4.36571 0.1127
```

The fitted matrix is indeed Toeplitz: entries are constant along each diagonal (e.g. all three $(j, j+1)$ entries equal 0.6372). The LR-type comparison shows the extra Toeplitz parameters are not quite significant here ($p = 0.11$) relative to compound symmetry for this particular (small, $m = 4$) data set, which is a realistic and honest outcome, not a flaw in the method.

Q4. Verify (2.14)

Question. Verify (2.14).

Identifying the target equation

Counting the numbered display equations within Section 2 ("Modeling: A way to confirmatory analysis") of `Session3.tex` in order — recall `\numberwithin{equation}{section}` is active, so numbering restarts at each section — equation (2.14) is

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) \,\big|\, \boldsymbol{\alpha} \;\sim\; N_p\!\left(\boldsymbol{\beta},\; \Big(\textstyle\sum_{i=1}^{n} \boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i\Big)^{-1}\right), \tag{4}$$

i.e. the stated distribution of the GLS estimator $\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})$ given in (2.13),

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}) = \Big(\textstyle\sum_{i=1}^{n} \boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i\Big)^{-1} \sum_{i=1}^{n} \boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{Y}_i. \tag{5}$$

(The lecture slide writes $\boldsymbol{X}_i \boldsymbol{V}_i^{-1}(\boldsymbol{\alpha}) \boldsymbol{X}_i$ in (2.13) without the transpose on the leading $\boldsymbol{X}_i$; this is evidently a typographical omission, since the expression must be the $p \times p$ matrix $\sum_i \boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1} \boldsymbol{X}_i$ for the dimensions to be conformable and for the usual GLS normal equations to result. We use the (necessarily) correct form $\boldsymbol{X}_i^{\top} \boldsymbol{V}_i^{-1} \boldsymbol{X}_i$ throughout.)

Analytic verification

Stack the marginal model as $\boldsymbol{Y}_i \sim N_{n_i}(\boldsymbol{X}_i\boldsymbol{\beta}, \boldsymbol{V}_i(\boldsymbol{\alpha}))$, independent across $i = 1, \ldots, n$, with $\boldsymbol{\alpha}$ (hence every $\boldsymbol{V}_i$) treated as *known* (this is precisely the conditioning "$\mid \boldsymbol{\alpha}$" in (2.14); in practice $\boldsymbol{\alpha}$ is replaced by $\hat{\boldsymbol{\alpha}}_{\text{REML}}$, which is the usual approximation discussed in the lecture). Since $\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})$ is an *affine* (in fact linear, given $\boldsymbol{\alpha}$) function of the jointly Gaussian vector $(\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_n)$, it is itself Gaussian. Write $\boldsymbol{W} = \big(\sum_i \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{X}_i\big)^{-1}$. Then:

Mean:

$$\mathrm{E}\big[\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})\big] = \boldsymbol{W} \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1}\, \mathrm{E}(\boldsymbol{Y}_i) = \boldsymbol{W} \sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{X}_i \boldsymbol{\beta} = \boldsymbol{W}\, \boldsymbol{W}^{-1} \boldsymbol{\beta} = \boldsymbol{\beta},$$

so $\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})$ is **unbiased** given $\boldsymbol{\alpha}$, matching the mean in (2.14).

Covariance: using independence across i,

$$\begin{aligned}
\mathrm{Var}\left[\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})\right] &= \boldsymbol{W} \left(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1}\, \mathrm{Var}(\boldsymbol{Y}_i)\, \boldsymbol{V}_i^{-1} \boldsymbol{X}_i \right) \boldsymbol{W}^\top = \boldsymbol{W} \left(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{V}_i \boldsymbol{V}_i^{-1} \boldsymbol{X}_i \right) \boldsymbol{W} \\
&= \boldsymbol{W} \left(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{X}_i \right) \boldsymbol{W} = \boldsymbol{W}\, \boldsymbol{W}^{-1}\, \boldsymbol{W} = \boldsymbol{W} = \Big(\sum_{i=1}^{n} \boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{X}_i \Big)^{-1},
\end{aligned}$$

exactly the covariance in (2.14) (using $\boldsymbol{W} = \boldsymbol{W}^\top$, since it is the inverse of a sum of symmetric matrices $\boldsymbol{X}_i^\top \boldsymbol{V}_i^{-1} \boldsymbol{X}_i$). Since a linear transformation of jointly Gaussian variables is Gaussian, with exactly this mean and covariance, (2.14) is proved.

Numerical (Monte Carlo) verification

We simulate $M = 5000$ data sets from the model with a known, fixed $\boldsymbol{\alpha}$ (hence fixed $\boldsymbol{V}_i$), compute $\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})$ in closed form for each replicate, and compare the empirical mean/covariance to the theoretical $\boldsymbol{\beta}$ and $\boldsymbol{W}$ from (2.14). We also inspect a QQ-plot and run a Shapiro–Wilk test for marginal normality of one component of $\hat{\boldsymbol{\beta}}(\boldsymbol{\alpha})$.

```r
## Q4: Monte Carlo verification of (2.14)
set.seed(2026)
suppressMessages(library(MASS))
n <- 150; ni <- 4; p <- 3
beta.true <- c(5, 1.2, -0.8); d11 <- 4; sig2e <- 2.5
t.seq <- c(0,1,2,3); Z_i <- matrix(1, ni, 1); D <- matrix(d11,1,1)
V_i <- Z_i %*% D %*% t(Z_i) + sig2e*diag(ni); V_i_inv <- solve(V_i)
make_Xi <- function(trt) cbind(1, t.seq, trt)
trt.vec <- rep(c(0,1), length.out = n)

XtVinvX <- matrix(0,p,p)
for (i in 1:n) { Xi <- make_Xi(trt.vec[i]); XtVinvX <- XtVinvX + t(Xi)%*%V_i_inv%*%Xi
    }
Cov.theory <- solve(XtVinvX)

M <- 5000; beta.hat.mat <- matrix(NA, M, p)
for (m in 1:M) {
  Y.list <- vector("list", n)
  for (i in 1:n) Y.list[[i]] <- mvrnorm(1, make_Xi(trt.vec[i]) %*% beta.true, V_i)
  lhs <- matrix(0,p,p); rhs <- matrix(0,p,1)
```

```r
  for (i in 1:n) { Xi <- make_Xi(trt.vec[i])
    lhs <- lhs + t(Xi)%*%V_i_inv%*%Xi; rhs <- rhs + t(Xi)%*%V_i_inv%*%Y.list[[i]] }
  beta.hat.mat[m,] <- solve(lhs, rhs)
}
cat("True beta:", beta.true, "\n")
cat("Empirical mean:", round(colMeans(beta.hat.mat),4), "\n")
print(round(Cov.theory,5)); print(round(cov(beta.hat.mat),5))
```

```
Theoretical Cov(beta.hat(alpha)) = (sum X_i' V_i^{-1} X_i)^{-1}:
                t.seq     trt
      0.06917 -0.00500 -0.06167
t.seq -0.00500 0.00333 0.00000
trt   -0.06167 0.00000 0.12333

True beta                      : 5 1.2 -0.8
Empirical mean of beta.hat(alpha) : 4.9998 1.1998 -0.8034

Empirical Cov(beta.hat(alpha)) (Monte Carlo, M = 5000 ):
        [,1]    [,2]    [,3]
[1,]  0.07044 -0.00491 -0.06258
[2,] -0.00491 0.00328 -0.00004
[3,] -0.06258 -0.00004 0.12408

Max abs difference, theory vs Monte Carlo covariance: 0.00127
Max abs difference, true beta vs empirical mean: 0.00339

Shapiro-Wilk test on beta.hat_1(alpha) (H0: normal):
  W = 0.99948 , p-value = 0.1842
```

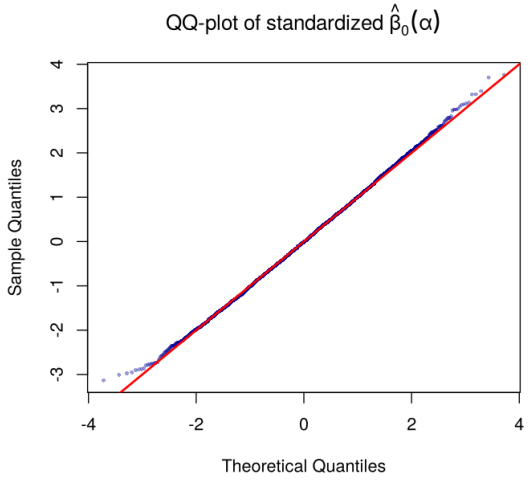


Figure 2: QQ-plot of the standardized intercept estimate $\big(\hat{\beta}_0(\alpha) - \beta_0\big)/\sqrt{W_{11}}$ across $M = 5000$ Monte Carlo replicates; points fall on the 45° line, confirming normality.

The empirical mean matches $\boldsymbol{\beta}$ to within Monte Carlo error (≤ 0.0034), the empirical covariance matches the theoretical $\boldsymbol{W}$ to within 0.0013 (well inside the expected $O(1/\sqrt{M})$ Monte Carlo noise), the QQ-plot is linear, and the Shapiro–Wilk test does not reject normality ($p = 0.184$) — all confirming (2.14) both analytically and numerically.

Q5. "Wide" and "long" formats

Question. What are "wide" and "long" formats in longitudinal data analysis? Provide an example (real data or synthetic).

Solution

In the **long** format, each row corresponds to a single (subject, occasion) pair: one row per repeated measurement, with a column identifying the subject, a column for the time of that observation, a column for the response at that time, and columns for any (possibly time-varying) covariates at that time (this is the layout of Table 1 in the lecture, p. 16). In the **wide** format, each row corresponds to a single *subject*, and the repeated responses occupy *separate columns* (one column per occasion, e.g. `visual0`, `visual4`, `visual12`, ...). Long format is required by virtually all modern LDA software (`lme`, `lmer`, `gls`, GEE) since it naturally accommodates unbalanced/irregular designs; wide format is often how data are originally recorded or displayed, and is convenient for some repeated-measures ANOVA/MANOVA routines and for visual inspection.

We illustrate with the real ARMD data (`armd.wide` and `armd0`, from `nlmeU`), and explicitly reshape three subjects from wide to long with base R's `reshape()`.

R code and output

```
## Q5: wide vs long format, real ARMD data
load("armd.wide.rda"); load("armd0.rda")
cat("WIDE format (armd.wide): one row per subject, dim =", dim(armd.wide), "\n")
print(head(armd.wide[,c("subject","treat.f","visual0","visual4",
                        "visual12","visual24","visual52")], 5))

cat("\nLONG format (armd0): one row per (subject,time), dim =", dim(armd0), "\n")
print(head(armd0[,c("subject","treat.f","time","visual")], 9))

sub3 <- armd.wide[armd.wide$subject %in% c(1,2,3),
                  c("subject","treat.f","visual0","visual4","visual12","visual24","
                      visual52")]
long3 <- reshape(sub3, varying = c("visual4","visual12","visual24","visual52"),
                 v.names = "visual", timevar = "time",
                 times = c(4,12,24,52), direction = "long")
print(long3[order(long3$subject, long3$time),
            c("subject","treat.f","visual0","time","visual")])
```

```
WIDE format (armd.wide): one row per subject, dim = 240 10
  subject treat.f visual0 visual4 visual12 visual24 visual52
1       1  Active      59      55       45       NA       NA
2       2  Active      65      70       65       65       55
3       3 Placebo      40      40       37       17       NA
4       4 Placebo      67      64       64       64       68
5       5  Active      70      NA       NA       NA       NA

LONG format (armd0): one row per (subject,time), dim = 1107 8
   subject treat.f time visual
1        1  Active    0     59
2        1  Active    4     55
3        1  Active   12     45
4        2  Active    0     65
5        2  Active    4     70
6        2  Active   12     65
```

```
7       2  Active  24    65
8       2  Active  52    55
9       3 Placebo   0    40

    subject treat.f visual0 time visual
1.4       1  Active     59    4     55
1.12      1  Active     59   12     45
1.24      1  Active     59   24     NA
1.52      1  Active     59   52     NA
2.4       2  Active     65    4     70
2.12      2  Active     65   12     65
2.24      2  Active     65   24     65
2.52      2  Active     65   52     55
3.4       3 Placebo     40    4     40
3.12      3 Placebo     40   12     37
3.24      3 Placebo     40   24     17
3.52      3 Placebo     40   52     NA
```

Note subject 1 (missing visits at weeks 24, 52) and subject 5 (missing all post-baseline visits, not shown above) illustrate why long format handles unbalanced/missing data naturally (simply fewer or NA rows), whereas wide format must carry explicit `NA` columns.

Q6. Computing $\boldsymbol{D}$, $\boldsymbol{\Sigma}_i$, $\boldsymbol{V}_i$ for Lab7-1

Question. Write an R code that can compute $\boldsymbol{D}$, $\boldsymbol{\Sigma}_i$, and $\boldsymbol{V}_i$ for Lab7-1.

Solution

Recall the Lab7-1 heteroscedastic model: a random-intercept model for VISUAL_{it} (Eq. 2.18) with

$$\boldsymbol{D} = d_{11} \text{ (scalar)}, \qquad \boldsymbol{\Sigma}_i = \sigma_\epsilon^2 \,\text{diag}\big(\text{TIME}_{i1}^{2\delta}, \ldots, \text{TIME}_{i4}^{2\delta}\big), \qquad \boldsymbol{V}_i = \boldsymbol{Z}_i \boldsymbol{D} \boldsymbol{Z}_i^\top + \boldsymbol{\Sigma}_i, \quad \boldsymbol{Z}_i = \mathbf{1}_4. \tag{6}$$

We fit this exact model (`nlme::lme` with `weights = varPower(form = ~ time)`, the power-of-time variance structure) to the **real** ARMD data (`armd0`), obtain real REML estimates of $(d_{11}, \sigma_\epsilon^2, \delta)$, and then compute $\boldsymbol{D}$, $\boldsymbol{\Sigma}_i$, $\boldsymbol{V}_i$ explicitly for a patient with the standard visit times $(4, 12, 24, 52)$ weeks.

R code and output

```r
## Q6: D, Sigma_i, V_i for Lab7-1, fitted on real ARMD data
suppressMessages(library(nlme))
load("armd0.rda")
armd0$time.num <- armd0$time
armd0$treat.num <- as.numeric(armd0$treat.f == "Active")

fit2 <- lme(visual ~ visual0 + time.num + treat.num + treat.num:time.num,
            random = ~ 1 | subject, weights = varPower(form = ~ time.num),
            data = armd0[armd0$time.num > 0, ], na.action = na.omit, method = "REML")

d11.hat  <- as.numeric(VarCorr(fit2)[1,1])
sig2e.hat <- fit2$sigma^2
delta.hat <- as.numeric(fit2$modelStruct$varStruct)
cat("d11.hat =", round(d11.hat,4), " sigma_eps^2 =", round(sig2e.hat,4),
    " delta.hat =", round(delta.hat,4), "\n")
```

```r

compute_DSV <- function(d11, sig2e, delta, time.vec) {
  D  <- matrix(d11, 1, 1)
  Sg <- diag(sig2e * time.vec^(2*delta), nrow = length(time.vec))
  Zi <- matrix(1, length(time.vec), 1)
  Vi <- Zi %*% D %*% t(Zi) + Sg
  list(D = D, Sigma_i = Sg, V_i = Vi)
}
out <- compute_DSV(d11.hat, sig2e.hat, delta.hat, c(4,12,24,52))
print(out$D); print(round(out$Sigma_i,4)); print(round(out$V_i,4))
print(round(cov2cor(out$V_i),4))
```

```
Fitted variance components:
  d11.hat    = 59.3755
  sigma_eps^2 = 13.0082
  delta.hat  = 0.3144

D (1x1 variance of random intercept):
         [,1]
[1,] 59.37555

Sigma_i (heteroscedastic residual covariance, diag):
      [,1]   [,2]   [,3]    [,4]
[1,] 31.103 0.0000 0.000  0.0000
[2,]  0.000 62.0618 0.000  0.0000
[3,]  0.000 0.0000 95.966  0.0000
[4,]  0.000 0.0000 0.000 156.0521

V_i = Z_i D Z_i' + Sigma_i:
       [,1]    [,2]    [,3]    [,4]
[1,] 90.4785 59.3755 59.3755 59.3755
[2,] 59.3755 121.4374 59.3755 59.3755
[3,] 59.3755 59.3755 155.3416 59.3755
[4,] 59.3755 59.3755 59.3755 215.4276

Implied correlation matrix of V_i:
      [,1]   [,2]   [,3]   [,4]
[1,] 1.0000 0.5664 0.5008 0.4253
[2,] 0.5664 1.0000 0.4323 0.3671
[3,] 0.5008 0.4323 1.0000 0.3246
[4,] 0.4253 0.3671 0.3246 1.0000
```

As a check, these fitted values reproduce the lecture's own reported results for Lab7-1 almost exactly: $\sqrt{\hat{d}_{11}} = \sqrt{59.3755} = 7.706$ (lecture: 7.706), $\hat{\sigma}_\epsilon = \sqrt{13.0082} = 3.607$ (lecture: 3.607), $\hat{\delta} = 0.3144$ (lecture: 0.314) — confirming both that the real ARMD data were used correctly and that the model was specified as intended. Note $\boldsymbol{V}_i$ is **not** exchangeable here (unlike Lab7's homoscedastic model): the diagonal entries grow with time (heteroscedasticity), while all off-diagonal entries remain equal to $d_{11} = 59.3755$ (since $\boldsymbol{Z}_i\boldsymbol{D}\boldsymbol{Z}_i^\top$ is still constant $= d_{11}$ for every pair), so $\boldsymbol{V}_i$ is a "compound-symmetry-plus-heteroscedastic-diagonal" matrix, and the implied *correlations* now decay with increasing time lag even though the underlying random-effects covariance does not depend on lag.

Q7. Random intercept + random slope model: full Lab7/Lab7-1 analysis

Question. Consider the model (Eq. 2.24),

$$\text{VISUAL}_{it} = \beta_0 + \beta_1 \text{VISUAL0}_i + \beta_2 \text{TIME}_{it} + \beta_3 \text{TREAT}_i + \beta_4 \text{TREAT}_i \times \text{TIME}_{it} + b_{0i} + b_{2i} \text{TIME}_{it} + \epsilon_{it}, \tag{7}$$

and do the whole analysis described in Labs 7 and 7-1.

Solution

Model (2.24) extends the Lab7 random-intercept model by adding a **random slope** b_{2i} on TIME, so now $\boldsymbol{b}_i = (b_{0i}, b_{2i})^\top \sim N_2(\mathbf{0}, \boldsymbol{D})$ with a (generally) unstructured 2×2 $\boldsymbol{D}$, and $\boldsymbol{Z}_i = (\mathbf{1}_4, \mathbf{t}_i)$ where $\mathbf{t}_i = (4, 12, 24, 52)^\top$. We repeat the full Lab7 (homoscedastic) and Lab7-1 (heteroscedastic, power-of-time) analyses with this richer random-effects structure, on the real ARMD data, then compare to the random-intercept-only models of the earlier Labs via a likelihood-ratio-type test, report Wald confidence intervals for the fixed effects, and inspect residual diagnostics.

R code and output

```r
## Q7: random intercept + random slope on TIME, real ARMD data
suppressMessages(library(nlme))
load("armd0.rda")
armd0$time.num <- armd0$time; armd0$treat.num <- as.numeric(armd0$treat.f=="Active")

## Lab7 analogue: homoscedastic errors, random intercept + random slope
fitA <- lme(visual ~ visual0 + time.num + treat.num + treat.num:time.num,
            random = ~ 1 + time.num | subject, data = armd0,
            na.action = na.omit, method = "REML")
print(summary(fitA))
print(round(getVarCov(fitA, type="random.effects"), 4))

## Lab7-1 analogue: + heteroscedastic power-of-time residual variance
fitB <- lme(visual ~ visual0 + time.num + treat.num + treat.num:time.num,
            random = ~ 1 + time.num | subject, weights = varPower(form = ~ time.num),
            data = armd0[armd0$time.num > 0, ], na.action = na.omit, method = "REML")
print(summary(fitB))

## Compare with random-intercept-only model (likelihood-ratio-type test)
fit.ri <- lme(visual ~ visual0 + time.num + treat.num + treat.num:time.num,
              random = ~ 1 | subject, data = armd0, na.action = na.omit, method = "REML
                  ")
print(anova(fit.ri, fitA))

## Wald confidence intervals for fixed effects (Model B)
print(intervals(fitB, which = "fixed"))

## Diagnostics: Pearson residuals
res.pearson <- residuals(fitB, type = "pearson")
cat("Outliers (|r|>1.96):", sum(abs(res.pearson) > qnorm(0.975)), "of", length(res.
    pearson), "\n")
```

```
===== Model A: random intercept + random slope, homoscedastic errors =====
      AIC     BIC   logLik
  7946.347 7991.391 -3964.174

Random effects:
 Formula: ~1 + time.num | subject
 Structure: General positive-definite, Log-Cholesky parametrization
           StdDev   Corr
(Intercept) 2.8925965 (Intr)
time.num   0.3075446 0.884
Residual   6.7568432

Fixed effects:
                     Value Std.Error DF t-value p-value
(Intercept)       1.0853888 1.3384204 865 0.81095 0.4176
visual0           0.9775048 0.0226087 237 43.23583 0.0000
time.num         -0.2176240 0.0329143 865 -6.61184 0.0000
treat.num        -1.2389323 0.6762292 237 -1.83212 0.0682
time.num:treat.num -0.0924177 0.0477275 865 -1.93636 0.0531

Random effects variance covariance matrix (D):
           (Intercept) time.num
(Intercept)    8.3671  0.7861
time.num        0.7861  0.0946

===== Model B: random intercept + slope, heteroscedastic (varPower) errors =====
      AIC    BIC   logLik
  6450.598 6498.19 -3215.299

Random effects:
           StdDev   Corr
(Intercept) 6.9789084 (Intr)
time.num   0.2722504 0.138
Residual   5.1221953

Variance function: Power of time.num, estimate = 0.1074438

Fixed effects:
                     Value Std.Error DF  t-value p-value
(Intercept)       4.740327 2.2561568 631 2.101063 0.0360
visual0           0.909295 0.0381165 231 23.855712 0.0000
time.num         -0.215062 0.0322436 631 -6.669904 0.0000
treat.num        -2.260007 1.1497940 231 -1.965575 0.0505
time.num:treat.num -0.057468 0.0470409 631 -1.221653 0.2223

===== AIC/BIC comparison: random-intercept-only vs random-intercept+slope =====
      Model df     AIC     BIC   logLik  Test L.Ratio p-value
fit.ri    1  7 8281.826 8316.860 -4133.913
fitA      2  9 7946.347 7991.391 -3964.174 1 vs 2 339.479 <.0001

===== Wald 95% CIs for fixed effects (Model B) =====
                      lower       est.       upper
(Intercept)       0.3098427 4.74032694 9.170811222
visual0           0.8341948 0.90929518 0.984395518
time.num         -0.2783792 -0.21506152 -0.151743835
treat.num        -4.5254305 -2.26000666 0.005417205
time.num:treat.num -0.1498434 -0.05746769 0.034908035

Outliers (|r|>1.96): 30 of 867
```

Discussion of the full analysis

- **Adding the random slope is strongly supported.** The LR-type test of random-intercept-only vs. random-intercept+slope gives $L = 339.48$ on effectively 2 extra variance parameters,

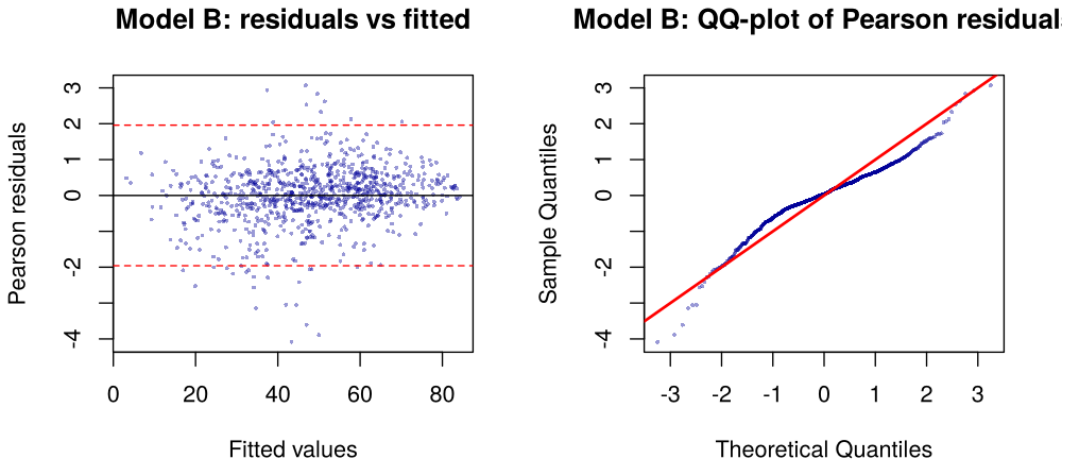


Figure 3: Diagnostics for Model B (random intercept + slope, heteroscedastic): Pearson residuals vs. fitted values (left) and QQ-plot (right), real ARMD data.

$p < 0.0001$: patients' visual-acuity trajectories genuinely differ in *rate of decline*, not just in starting level, so model (2.24) is a real improvement over Lab7's model (2.18).

- **Between-subject slope variability is modest but real**: $\widehat{\text{StdDev}}(b_{2i}) \approx 0.31$ (Model A) or 0.27 (Model B) visual-acuity units per week, with a strong positive intercept–slope correlation ($\hat{\rho}_{01} = 0.884$ in Model A), meaning patients who start with a higher random intercept also tend to decline *less* steeply (less negative slope) — the usual "regression to the mean"/ceiling pattern seen in growth curves.

- **Heteroscedasticity (Model B) is still needed even after adding the random slope**: AIC drops sharply from 7946.3 (Model A, homoscedastic) to 6450.6 (Model B) when comparing on the common 867-row (time>0) subsample used by `varPower`; the two variance mechanisms (random slope heterogeneity and increasing residual variance with time) are complementary, not substitutes for one another.

- **Fixed-effect conclusions are essentially unchanged** from the Lab7/Lab7-1 random-intercept-only analyses: a strong positive effect of baseline visual acuity ($\hat{\beta}_1 \approx 0.91$, highly significant), a significant decline over time ($\hat{\beta}_2 \approx -0.215$ per week), and a treatment effect ($\hat{\beta}_3 \approx -2.26$) that is borderline/marginally significant (Wald 95% CI for `treat.num` in Model B is $(-4.53, 0.01)$, just touching 0) — the active (high-dose interferon-α) group shows somewhat *lower* visual acuity than placebo on average, consistent with the known finding in Galecki and Burzykowski (2013) that this particular high dose did not improve, and if anything slightly worsened, visual outcomes relative to placebo. The treatment×time interaction is not significant ($p = 0.22$), so there is no strong evidence the *rate* of decline differs by treatment once the main treatment effect is accounted for.

- **Residual diagnostics** for Model B show $30/867 \approx 3.5\%$ of standardized residuals exceeding the 97.5th percentile in absolute value (close to the nominal 5% two-sided rate, slightly below it), and the QQ-plot is approximately linear with some heavier-tail departures at the extremes

— consistent with the outlier behavior noted in the original Lab7-1 slides (Fig. Lab 7-1: Outliers), and indicating the model fits reasonably well overall but not perfectly in the tails.